

RISHI SHAH

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Education

McMaster University

Apr. 2029

Engineering - Schulich Leader Scholar (1 of 50 in Canada to win \$120K national STEM scholarship)

Hamilton, Ontario

Experience

The Centre for Mechatronics and Hybrid Technologies

Oct. 2025 – Present

Machine Learning Researcher

Hamilton, ON

- Engineered a hybrid LSTM model on top of PyBaMM SPMe simulations to predict anode potential, reducing RMSE from 8 mV to 1.24 mV
- Built hybrid SPMe-ML voltage-correction model improving SPMe RMSE from 56.5 mV to 2.6 mV

Western University

Apr. 2024 – Aug. 2025

Machine Learning Researcher

London, ON

- Engineered a recurrent, denoising CORNet-S variant that outperformed MIT's CORNet-S by up to 10% in adversarial robustness across MNIST, CIFAR-100, and ImageNet100
- Achieved 97.82% adversarial accuracy, surpassing ResNet-18 and AlexNet in PGD, CW, and patch-attack benchmarks

3D Forge

May 2024 – Sep. 2024

Founder

London, ON

- Scaled a 3D-printing business to 100+ customers and \$3,500+ profit in one summer
- Awarded Ontario's Summer Company Grant for student entrepreneurship

Robarts Research Institute

Nov. 2022 – May 2024

Student Researcher

London, ON

- Developed mixed-reality AR visualization systems to model progression/recovery of arterial disease over time
- Designed and trained CNN-based segmentation models on DICOM images
- Implemented 3D ultrasonic tracking pipelines for spatial localization of surgical instruments

Projects

Adaptive Learning Robot | C++, Arduino, Python, Cohere LLM | Top 32 at Hack the North

Sept. 2025

- Built a dual-system robot inspired by Kahneman's System 1 / System 2 model that learns new skills instantly from natural language without pre-training or rigid scripts
- Implemented Cohere LLMs with real-time search to convert messy instructions into smooth, executable motions for 2 custom 3-DOF robotic arms

Sentinel | C++, TypeScript, JavaScript, Node.js | Best Developer Tool at HackWestern 12

Nov. 2025

- Built a hardware-aware static analyzer with a custom TypeScript LSP performing 50+ silicon-level checks, including RAM/Flash budgeting, I²C/SPI conflict detection, and pin-map validation.
- Developed full VS Code/Cursor integration for real-time diagnostics on embedded C/C++ projects

From Pixels to Precision | Python, MatLab | National Bronze; Divisional Gold; Sanofi BioGenius Award

May 2023

- Partnered with Synaptive Medical Inc. and surgeons to develop a deep-learning CNN to track surgical tool movements in minimally invasive surgeries
- Trained on 300 manually segmented surgical images and achieved 95% DICE accuracy
- Analyzed performance across 4 illumination and tool scenarios with sub-0.12s inference speed per image

Leadership & Extracurricular Activities

Hack49 Global

Jun. 2024 – Jun. 2025

Co-founder and Lead of Outreach

Remote

- Built a global programming community of 950+ students across 40+ countries with industry recognition from TD, Spotify, and Amazon
- Secured \$19,000+ in sponsorships and partnered with Elegoo

DECA Chapter Lead

Sep. 2023 – Jun. 2025

President

London, ON

- Mentored 80+ students, and grew chapter by 55%, raising provincial qualifiers by 52% and international by 50%

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, Java, C#, MATLAB, SQL, Bash

Frameworks: PyTorch, Keras, OpenCV, YOLO, React, Django, Flask, TailwindCSS, Electron, Node.js

Tools: Arduino, Raspberry Pi, ESP32, Unity, Git, VSCode, AWS, Fusion 360, Blender, OnShape, KiCad, Docker, Linux